

Polymarket-Conditioned Equity Returns: A Polarity-Aware Empirical Study

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Abstract

We test whether equities linked to Polymarket questions exhibit positive average net return after a probability-based eligibility signal and before the associated event deadline. The object of interest is a conditional expectation, not a causal information-diffusion mechanism. The raw archive contains 1,293 question-asset candidate rows from August 2024 through June 2026. Exact duplicate collapse and a targeted semantic audit produce a primary universe of 1,182 candidates with explicit contract polarity. A fixed-notional H1 evaluation yields 887 tradable observations with mean net return +1.208% and median +0.188%. The candidate mean remains positive when resampling economic events, but confidence intervals cross zero under symbol and calendar dependence, and no interval excludes zero once each trade is compared against a matched, equal-notional SPY or sector-ETF position over the identical dates. Equal-weight economic events have mean +0.631%, and removing the largest one percent of events reduces this estimate to +0.029%. Geopolitical observations carry the result but are concentrated in oil-linked instruments and one month. We then evaluate a secondary Cross-Entropy Method (CEM) portfolio implementation with four treatments: friction-aware fitness, label-complete walk-forward fitting, bounded half-Kelly sizing, and event-priority allocation. Across ten seeds and four CEM budgets, median excess return over the single January–June 2026 test period is positive on both benchmarks for every budget; however, in seed-paired comparisons the treated configuration does not outperform the frozen batch-fit baseline. The evidence supports Polymarket probabilities as continuous, timestamped crowd-belief variables for candidate identification; it does not establish market-relative excess return, identify why the conditional return exists, or support a production-level profitability claim.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Model development and analysis*.

Keywords

prediction markets, event-driven equities, conditional expectation, semantic polarity, walk-forward backtesting

1 Introduction

Prediction markets provide continuously revised prices for discrete future events. Unlike a conventional news archive, a market price supplies a time-indexed state variable: the current crowd-implied probability of a contract resolving YES. This creates a practical link between event identification and equity timing when a contract can be mapped to an economically exposed asset. The link is not automatic. Contract wording, outcome direction, event deadlines,

duplicated questions, and non-equity resolution rules must be interpreted before the probability can be used as a bullish or bearish signal.

This paper studies the following empirical question:

Within a Polymarket-identified and semantically cleaned opportunity set, is the average net equity return positive between probability-based signal eligibility and the final eligible pre-event close?

The associated estimand is deliberately narrower than abnormal return, alpha, or slow information diffusion. It asks whether the observed conditional mean is positive in the opportunity set generated by the system. A positive estimate can arise from several mechanisms, including risk compensation, common-news exposure, event selection, or delayed incorporation of information. Mechanism identification is outside the present test.

The paper makes five empirical contributions: explicit contract polarity, so a favorable NO outcome is not mechanically read as bearish; deterministic duplicate collapse plus a targeted semantic audit before estimation; expectation estimates at candidate, symbol-day, and economic-event levels with cluster and block resampling, and separately against matched equal-notional benchmark trades; exact eligibility-rule provenance — the entry thresholds are outputs of the portfolio optimizer’s walk-forward schedule, so the full-sample H1 estimate is descriptive rather than an independent confirmatory test; and one portfolio implementation evaluated across ten random seeds and four CEM budgets, including seed-paired treatment-versus-baseline comparisons. The H1 estimate and the CEM portfolio are kept separate: H1 describes the opportunity set; CEM tests whether a constrained implementation can use it.

2 Related Work

Prediction-market prices are commonly interpreted as aggregators of dispersed beliefs [1, 4]. Their empirical forecasting value is documented in both public and corporate settings [5, 7]. However, market prices need not be calibrated probabilities in every contract: liquidity, participant composition, resolution wording, and correlated information sources affect interpretation.

The efficient-market benchmark predicts rapid incorporation of public information into asset prices [2], while costly information acquisition and trading frictions permit incomplete adjustment [3]. Delayed equity responses are documented in settings such as post-earnings-announcement drift [8], and slow-moving capital supplies one possible frictional channel [6]. These literatures motivate the empirical design, but the present data do not distinguish delayed diffusion from alternative explanations. We therefore treat Polymarket as a continuous belief measurement system rather than assume a particular causal mechanism.

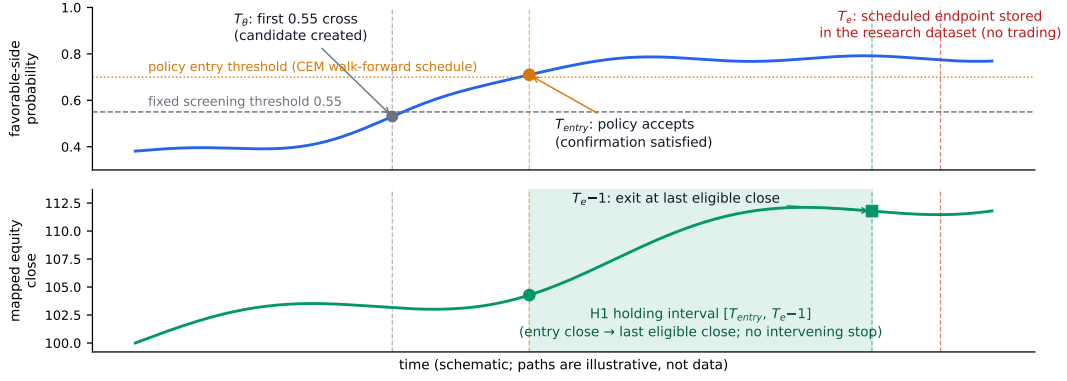


Figure 1: The H1 estimand and its chronology. T_θ is the fixed 0.55 screening cross that creates the candidate; T_{entry} is the first stored close at which the walk-forward eligibility policy accepts it. H1 measures the mapped equity from the T_{entry} close through the last eligible close before the scheduled endpoint T_e stored in the research dataset; it does not impose an intervening take-profit or stop-loss. The paths are schematic and not data.

3 Data and Candidate Construction

3.1 Observation unit and chronology

Each source row links one binary Polymarket contract to one U.S.-listed equity or exchange-traded fund. The stored data include the contract question, market and event identifiers, probability path, mapped symbol, scheduled endpoint, daily equity closes, and features available when the candidate is evaluated. The archive spans August 2024 through June 2026. Decisions are daily because a complete historical hourly equity panel is unavailable for every symbol.

Let $p_{i,t}$ denote the probability assigned to the economically favorable side of candidate i . Screening time $T_{\theta,i}$ is the first time this probability crosses a fixed 0.55 screening threshold; it is set at candidate construction and is not optimized. Entry acceptance is a separate, generally later event governed by the eligibility policy of Section 4.2: the favorable probability must reach a strong threshold immediately, or hold above a floor threshold for a required number of consecutive observations, with caps on the prior probability surge and on the pre-entry equity run-up. The entry time $T_{\text{entry},i}$ is the first stored equity close on or after acceptance. The endpoint $T_{e,i}$ is the scheduled contract endpoint stored in the research dataset. H1 exits at the final stored equity close strictly before that endpoint. The holding interval is therefore $[T_{\text{entry},i}, T_{e,i} - 1]$ rather than an early take-profit or stop-loss path, as summarized in Figure 1.

3.2 Candidate generation and relevance scoring

Candidates are constructed by a deterministic-then-LLM pipeline. Deterministic prefilters remove structural noise (leaderboards, rankings, word-count and price-level ladders) without scoring economic relevance, and exact duplicates are normalized before any model call. A Gemini-family language model (the exact version is configured at runtime) is then applied in two structured passes whose outputs are validated against fixed JSON schemas. The first pass rates question-level relevance $R_i^q \in [0, 1]$ — how mechanically a YES resolution transmits to U.S.-listed equities — and gates at $R_i^q \geq 0.60$, additionally requiring a positive-tone reading for the

long-only book. The second pass maps each surviving question to exposed U.S.-listed stocks or ETFs and grades a per-asset connection strength $C_{i,a} \in [0, 1]$. The stored pair relevance is $R_{i,a} = R_i^q C_{i,a}$, and all experiments require $R_{i,a} > 0.50$. Signal polarity (Section 3.3) is labeled in a separate pass and resolved by precedence: hand-audited override, then LLM label, then keyword fallback. A targeted human audit reviewed 109 question-asset pairs (Section 3.4); no accuracy claim is made for unaudited mappings.

3.3 Polarity

Raw Polymarket data record the probability of YES. That is insufficient for an equity mapping because YES is not universally favorable to the mapped asset. We assign

$$s_i \in \{-1, 0, +1\}, \quad p_{i,t}^+ = \begin{cases} p_{i,t}, & s_i = +1, \\ 1 - p_{i,t}, & s_i = -1, \end{cases}$$

where $s_i = +1$ denotes YES-bullish, $s_i = -1$ denotes NO-bullish, and $s_i = 0$ denotes no stable long-equity interpretation. Deterministic rules cover simple contract classes such as single-company regulatory approval. More complex questions use the audited semantic label. Polarity is an ontology decision: it states which outcome is favorable. It is not itself a learned return forecast.

The cleaned primary universe contains 1,071 YES-bullish candidates, 99 NO-bullish candidates, and 12 candidates without a usable side. Among valid H1 observations, 844 are YES-bullish and 43 are NO-bullish. This imbalance is important when interpreting the polarity ablation in Section 5.

3.4 Duplicate collapse and semantic audit

Exact source duplicates are removed globally before any return calculation. The targeted audit then examines 109 unique question-asset pairs selected from macroeconomic, geopolitical, regulatory, process, and non-standard contracts. Primary contracts are retained. Contracts that are circular, resolve from the mapped stock price, have unusable endpoints, or do not define an equity exposure are

Table 1: Candidate cleaning reconciliation.

Stage	Rows
Raw candidate rows	1,293
Exact duplicate excess removed	36
Rows after exact collapse	1,257
Audited unique question-asset pairs	109
Quarantined pending transformation	54
Secondary-only	11
Excluded or endpoint-invalid	10
Primary cleaned output	1,182

excluded. Process-only questions and contracts requiring an unavailable multi-contract probability transformation are marked secondary or quarantined rather than forced into H1.

Table 1 reconciles the data flow. The audit is targeted rather than universe-complete: 1,148 rows pass through deterministic rules without manual question-level review, and this boundary is recorded explicitly.

4 H1 Expectation Design

4.1 Estimand and execution

For each valid candidate, the primary net return is

$$r_i^{net} = \frac{P_{i,T_{e-1}} - P_{i,T_{entry}}}{P_{i,T_{entry}}} - \frac{C_i}{N_i},$$

where P is the stored equity close, $N_i = \$10,000$ is fixed entry notional, and C_i contains round-trip commissions, regulatory fees, and 5 basis points of slippage per transaction leg. Whole shares are used. Every candidate is evaluated independently; portfolio capital, concurrency, Kelly sizing, and event-priority allocation are absent from H1.

The primary hypothesis is $H_1 : \mu_O = \mathbb{E}[r_i^{net} \mid i \in O_{clean}] > 0$. The expectation is reported at three levels. Candidate observations retain every valid question-asset row. Symbol-day collapse retains one observation for the same symbol and entry day. Economic-event aggregation first combines sibling contracts and repeated assets belonging to the same underlying event, then weights events equally. The event view prevents a dense question family from being mistaken for independent evidence.

4.2 Eligibility provenance

The eligibility parameters (strong and floor entry thresholds, confirmation days, probability-surge cap, and run-up cap) are not fixed a priori. They are the walk-forward policy schedule fitted by the portfolio optimizer of Section 6 (the all-treatment SPY arm). Each fold’s policy is fitted only on candidates whose outcomes completed before that fold began, so every 2026 observation is evaluated under parameters that were point-in-time legal at its own signal date; however, the schedule is refit as the test period progresses, and observations before the first fold reuse the first fold’s policy retrospectively. Two consequences are stated plainly. First, the full-sample H1 estimate is descriptive evidence about the opportunity set under the system’s own selection rule, not an independent confirmatory hypothesis test. Second, the January–June 2026 subsample is the

closest the data come to out-of-sample: its eligibility decisions use no future outcomes, but its policies were refit within the period and the period itself has been repeatedly inspected during research.

4.3 Inference and sensitivity analysis

An iid one-sided t -test is shown only as a reference. The primary uncertainty analysis resamples economic-event clusters. Additional cluster or block schemes use symbol, entry week, and entry month. Each bootstrap uses 20,000 replications with fixed seeds. Reported p -values are null-centered and one-sided ($p = (\#\{\text{null mean} \geq \text{observed}\} + 1) / (B + 1)$); intervals are ordinary uncentered percentile intervals of the resampled mean. Medians and positive-return frequencies are reported because a right-skewed mean can be positive when barely half of observations win.

Sensitivity tests multiply modeled costs by up to three, shift entry to the next stored equity close, remove the largest positive observations, symmetrically trim both tails, and recompute estimates by calendar month and event family. A separate benchmark-relative analysis (Section 5.4) compares every trade against a matched benchmark position. These tests do not create new hypotheses; they measure how much of the central estimate is attributable to execution assumptions, shared calendar shocks, common market exposure, and the upper tail.

5 H1 Results

5.1 Central estimate

Of 1,182 cleaned candidates, 934 satisfy the probability entry rules. Eight hundred eighty-seven have a valid entry before the final eligible pre-event close; the other 47 reach acceptance only at or after that close and are excluded (no candidate is lost to missing price data). The rejected remainder comprises 121 candidates below the entry thresholds, 108 exceeding the probability-surge cap, 12 without a clean signal side, and 7 exceeding the run-up cap. Table 2 reports the central results. Candidate observations have mean net return +1.208%, median +0.188%, and positive-return frequency 51.97% across 750 economic events, 387 symbols, 46 entry weeks, and 16 entry months. The event-cluster bootstrap interval is [+0.263%, +2.200%] with one-sided $p = 0.0087$.

The result weakens as dependence becomes coarser. The candidate-symbol interval is [−0.184%, +2.776%] with $p = 0.0652$, the entry-week block interval is [−0.185%, +2.834%] with $p = 0.0678$, and the entry-month block interval is [−0.630%, +3.651%] with $p = 0.1427$. After equal-weight event aggregation, the mean is +0.631% and the median is +0.013%; its month-block interval is [−0.417%, +2.407%] with $p = 0.1935$. Figure 2 makes the dependence assumption visible rather than selecting one favorable standard error.

Restricting to the 528 candidates entered during January–June 2026 (point-in-time eligibility; Section 4.2) gives mean +2.038% and median +0.557%, with event-cluster $p = 0.0006$ but month-block $p = 0.1083$ on only six months.

5.2 Tail, cost, and timing sensitivity

The positive mean is upper-tail dependent. Removing the largest one percent of candidate returns lowers the candidate mean to +0.662%; removing the largest five percent makes it −0.260%. At

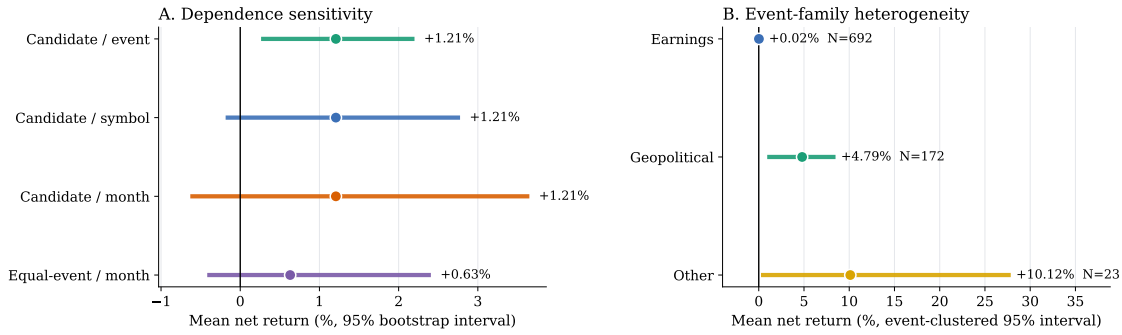


Figure 2: Cleaned H1 estimates. Panel A holds the candidate mean fixed while changing the resampling unit; the equal-weight event estimate is shown separately. Panel B is descriptive event-family heterogeneity, with candidate-weighted means and event-clustered intervals.

Table 2: Cleaned H1 expectation. All returns are net of modeled costs. Bootstrap p -values are null-centered and one-sided under the stated dependence unit; intervals are uncentered percentile intervals (20,000 replications).

Level and dependence unit	<i>N</i>	Mean	Median	Positive	95% bootstrap CI	p_{boot}
Candidate / economic event	887	+1.208%	+0.188%	51.97%	[+0.263, +2.200]%	0.0087
Candidate / symbol	887	+1.208%	+0.188%	51.97%	[-0.184, +2.776]%	0.0652
Candidate / entry month	887	+1.208%	+0.188%	51.97%	[-0.630, +3.651]%	0.1427
Symbol-day collapsed	786	+0.568%	-0.057%	49.62%	–	0.0611
Equal-weight event / entry month	750	+0.631%	+0.013%	50.27%	[-0.417, +2.407]%	0.1935

the event level, removing the largest one percent reduces the mean from +0.631% to +0.029% ($p = 0.4523$), and removing the largest five percent produces -0.715%. Symmetric trimming is less destructive: a symmetric five-percent trim leaves candidate mean +0.584% and event mean +0.070%. This pattern is consistent with a broadly dispersed but strongly right-skewed payoff distribution.

At three times modeled transaction costs, the candidate mean remains +0.954% ($p = 0.0026$), although its median becomes slightly negative. Repricing entry at the next stored equity close yields candidate mean +1.229% and event mean +0.550%; the event one-sided p -value is 0.0550. Thus the headline is not eliminated by a one-close delay. However, the stored daily data do not establish same-session ordering between the probability signal and the equity close (0 of 887 entries are verifiable), so the next-stored-close specification is reported as the conservative timing sensitivity.

5.3 Polarity and event-family heterogeneity

A controlled ablation treats every YES probability as bullish while leaving the cleaned candidate file and H1 policy unchanged. That version produces 882 observations with mean +1.272% and median +0.229%, compared with 887 observations, mean +1.208%, and median +0.188% under resolved polarity. Therefore polarity does not improve aggregate return in this historical sample. Its contribution is semantic validity: it prevents logically inverted signals and permits contracts with favorable NO outcomes to be represented correctly. The 43 selected NO-bullish observations have mean -2.100% and median -2.864%, so that subgroup requires prospective review rather than a performance claim.

Event-family results are more informative for candidate choice. Earnings observations ($N = 692$) have mean +0.020% with event-clustered interval [-0.476%, +0.521%]. Geopolitical observations ($N = 172$, only 37 economic events) have mean +4.794%, median +2.443%, positive-return frequency 61.63%, and event-clustered interval [+0.936%, +8.485%]. The residual other group has mean +10.118% but only 23 observations and a very wide interval. The family pattern is also concentrated: removing the single most influential symbol (the oil ETF USO, 104 rows) lowers the full-sample candidate mean from +1.208% to +0.313%, and removing March 2026 alone (116 rows) lowers it to +0.206% (at the event level, from +0.631% to +0.435%). The geopolitical result should therefore be read as substantially oil-linked and period-specific, not as a general property of geopolitical contracts. These subgroup results were not separately preregistered; they motivate family-aware allocation but do not establish that geopolitical classification is causal.

5.4 Benchmark-relative inference

The raw expectation could reflect common market exposure during a rising sample period. Each valid trade is therefore matched to an equal-notional benchmark trade over the identical entry and exit dates, with commissions, fees, and slippage modeled independently on both legs. Two controls are used: SPY for all 887 observations, and the candidate's actual sector ETF for the 680 observations with a real sector mapping (ETF-mapped geopolitical and macro candidates have no sector assignment and are excluded from the sector control rather than proxied by SPY).

Table 3 and Figure 3 report the result: the mean excess return is positive but no 95% interval excludes zero under any dependence

Table 3: Benchmark-relative H1 inference. Each stock trade is matched to an equal-notional benchmark trade over the identical entry and exit dates, with costs modeled independently on both legs. No interval excludes zero.

Benchmark	Level / unit	N	Mean (%)	95% CI (%)	p
SPY	Cand. / event	887	+0.91	[−0.12, +2.01]	0.051
SPY	Cand. / month	887	+0.91	[−0.90, +3.40]	0.204
SPY	Sym.-day / event	786	+0.22	[−0.45, +0.98]	0.264
SPY	Equal-event	750	+0.43	[−0.22, +1.17]	0.114
Sector ETF	Cand. / event	680	+0.19	[−0.39, +0.90]	0.264
Sector ETF	Cand. / month	680	+0.19	[−0.43, +1.46]	0.308
Sector ETF	Sym.-day / event	680	+0.19	[−0.39, +0.91]	0.266
Sector ETF	Equal-event	678	+0.18	[−0.40, +0.91]	0.273

scheme. Against SPY, the candidate mean excess is +0.914% with event-cluster $p = 0.0508$; symbol-day collapse reduces it to +0.223%, and the equal-weight event excess is +0.430% ($p = 0.1138$). Against sector ETFs the candidate excess is +0.191% ($p = 0.2644$). The H1 evidence is therefore a positive raw conditional mean whose market-relative component is not statistically established in this sample; a substantial part of the raw headline coincides with market and sector movement over the same holding windows.

6 CEM Portfolio Implementation

6.1 Purpose and baseline policy

H1 evaluates every eligible observation at fixed notional. The portfolio experiment asks a different question: can one constrained policy select and size overlapping candidates while capital remains invested in either event positions or the benchmark? Starting capital is \$100,000. Event positions use whole shares; idle cash above \$100 is swept into fractional SPY or QQQ holdings. The simulator applies Interactive Brokers-style commissions, SEC fees on sales, and 5 basis points of slippage on each trade leg; per-trade profit is net of all four rotation legs (benchmark sell, asset buy, asset sell, benchmark re-buy).

CEM searches ten bounded policy parameters: ATR exit multiple [1.5, 4.0], profit-lock activation [2%, 10%], probability exit threshold [0.45, 0.60], strong and floor entry thresholds, holding confirmation of one to five days, maximum probability surge [0.20, 0.80], maximum pre-entry equity run-up [2%, 20%], base position size [6%, 12%], and maximum concurrency of 8–12 positions. The baseline fitness is

$$J(\pi) = \text{Sharpe}_{\text{daily}}(\pi) - 0.30 | \text{MDD}_{\%}(\pi) |,$$

with the daily-equity Sharpe annualized by $\sqrt{252}$ and the maximum drawdown entering in percentage points. Each standard run uses six CEM iterations, population 20, elite fraction 25%, and a fixed seed. The robustness study varies both seed and CEM budget.

6.2 Treatments T1–T4

Each treatment changes one identifiable component of the implementation, following the portfolio branch in Figure 4. All treatment inputs are fixed constants or information available by the relevant decision time.

T1: realized-friction fitness. Let f_{fail} be the fraction of fitted trades whose gross PnL is below three times modeled transaction cost. T1 changes the CEM objective to $J(\pi) - 2f_{\text{fail}}$, preferring policies whose realized trades clear friction more consistently. It is not a live entry hurdle: the implementation has no point-in-time expected-return forecast, and T1 rejects no trade at execution time.

T2: label-complete walk-forward fitting. T2 constructs expanding three-month folds. A fold may fit only candidates with T_e strictly before the fold start; the following three-month candidate block is evaluated out of fold. At least two folds are required, with at least eight fit and eight evaluation candidates per fold. Every simulation is truncated at its fold horizon. Because the folds extend across the whole sample, T2 arms continue to refit inside the 2026 test period using only outcomes completed before each fold starts; they are online-refit schedules, not policies frozen at the end of 2025.

T3: bounded half-Kelly sizing. After at least ten completed net trades, T3 estimates win probability and the mean win/loss ratio from the latest 30 completed trades. It applies one-half Kelly and clips the resulting allocation to 5–20%. Before ten completed trades, or when both wins and losses are not observed, it uses the CEM base size. Open positions and future outcomes never enter the sizing history.

T4: event-priority allocation. T4 changes the allocation of limited concurrent slots. Same-day candidates are ranked by event family (geopolitical, macro, earnings, other), then by favorable entry probability and pre-entry equity run-up clipped to [−20%, +20%]. Same-symbol same-day signals are collapsed into one position. If every slot is occupied by earnings positions, a geopolitical or macro candidate may replace the worst earnings position only when that position is below +3% net. T4 changes portfolio allocation, not the H1 observation set.

6.3 Chronological evaluation and optimizer robustness

The cleaned candidate file contains 427 training rows dated August 2024 through December 2025 and 755 rows dated January through June 2026; after the relevance filter, 1,148 candidates enter the experiment, of which 721 fall in the test window. Training eligibility depends on label completion: a candidate can enter a CEM fit only when its event endpoint is strictly earlier than the fit cutoff. Portfolio simulation in every fold is also truncated at that fold’s decision horizon. No validation split is used anywhere in the pipeline: configuration families were declared in advance, and January–June 2026 is reported as one test period.

Two fitting protocols appear in the matrix, and both are leakage-free for the 2026 test. The Baseline and T4-alone arms fit once on the 423 candidates whose outcomes completed before 2026, with all simulations truncated at 2025-12-31, and stay *frozen* across the test; this is exactly the information a live operator would hold on 2026-01-01. Their training-path metrics are by construction retrospectively optimized (the policy is selected on the same history) and are not reported as performance. The T2 arms replace the frozen policy with the expanding walk-forward schedule, which keeps adapting during the test on completed outcomes only. The protocols differ

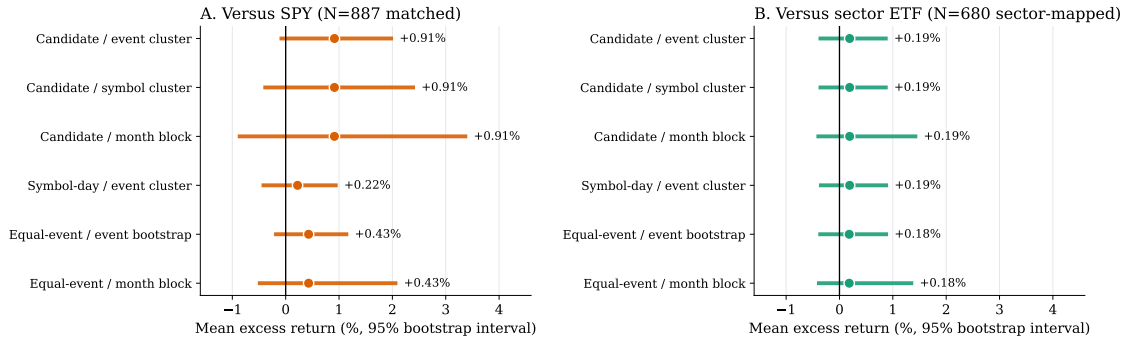


Figure 3: Benchmark-relative H1 inference: matched equal-notional benchmark trades over identical dates, after costs on both legs. Every 95% interval crosses zero.

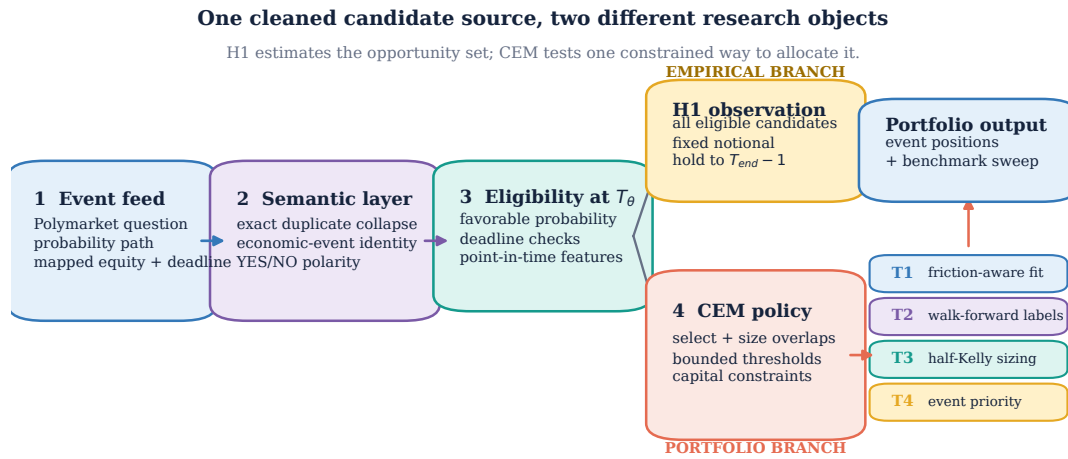


Figure 4: Research architecture. Cleaning and polarity create one eligible candidate source, after which the analysis deliberately separates two objects. H1 evaluates every eligible candidate at fixed notional through $T_{end} - 1$. The CEM branch selects and sizes overlapping candidates under capital constraints; T1–T4 modify fitness, chronology, sizing, and allocation, respectively. H1 output does not feed the portfolio results.

in whether the policy may adapt during the test, not in test-period validity.

The first result matrix uses seed 42 and the standard 6×20 budget for six configurations: the frozen batch-fit Baseline, T2 only, T1+T2, T1+T2+T3, T4 alone, and all four treatments. Because a single optimizer draw can be lucky, a second experiment evaluates the flagship all-treatment arm across seeds 42–51 and budgets (6, 20), (10, 20), (6, 30), and (10, 30), and pairs each seed’s all-treatment run against a Baseline run with the same seed at the standard budget. Distributions are summarized by median and interquartile range across seeds; the four budgets are robustness diagnostics, and no budget is selected or ranked by test performance.

7 CEM Results

7.1 Configuration matrix

Table 4 reports the seed-42, 6×20 matrix over the single January–June 2026 test period. All six configurations have positive absolute

return; excess return is positive everywhere except the T2-only arm on QQQ (−0.67). SPY excess ranges from +16.98 to +22.35 percentage points. The T2-only row isolates the fitting-protocol change: at this seed, replacing the frozen batch fit with walk-forward refitting costs −1.22 points on SPY and −10.44 points on QQQ before any other treatment is added; given T2, T1 adds back +1.82 and +7.70 points, T3 subtracts −5.37 and −2.05, and T4 adds +2.99 and +2.04. These are single-seed decompositions and are descriptive only. The absence of a common winner across benchmarks is evidence against selecting one configuration from this matrix by its best terminal return. Return, Sharpe, and drawdown values were recomputed from the daily equity logs and match the reported results.

7.2 Effect of candidate cleaning

To isolate the cleaning change, the all-treatment arm is rerun on the original and audit-cleaned candidate universes using identical current code, seed 42, and 6×20 budget. On SPY, terminal return rises

Table 4: Audit-cleaned CEM test matrix, January–June 2026 (seed 42, 6×20). Returns and excess are percentage points; buy-and-hold returned +8.52 (SPY) and +17.59 (QQQ). Baseline and T4 GeoPriority are frozen batch-fit policies; rows containing T2 are online walk-forward schedules (refits use completed outcomes only).

Configuration	SPY benchmark					QQQ benchmark				
	Return	Excess	MaxDD	Sharpe	Trades	Return	Excess	MaxDD	Sharpe	Trades
Baseline (frozen batch fit)	30.27	21.75	−6.67	3.30	225	27.36	9.77	−6.36	2.93	223
T2 only (walk-forward)	29.05	20.53	−10.20	3.05	197	16.92	−0.67	−8.67	1.79	191
T1+T2	30.87	22.35	−9.52	3.41	191	24.63	7.03	−7.51	2.52	189
T1+T2+T3	25.50	16.98	−7.76	2.99	197	22.57	4.98	−11.43	2.10	181
T4 GeoPriority	27.27	18.75	−8.33	2.89	197	23.58	5.99	−6.74	2.54	211
T1+T2+T3+T4	28.49	19.97	−8.72	3.22	184	24.61	7.02	−10.38	2.20	185

from 25.53% to 28.49%, a +2.96 point difference. On QQQ it rises from 22.29% to 24.61%, a +2.32 point difference. Five-day moving-block bootstrap tests of the cleaned-minus-original daily return give $p = 0.3371$ for SPY and $p = 0.3884$ for QQQ. Cleaning therefore improves these terminal point estimates, but the short matched path does not establish a systematic portfolio improvement.

7.3 Ten-seed and CEM-budget robustness

Figure 5 and Table 5 summarize 40 isolated flagship runs over the full test period. Median excess return is positive for every budget on both benchmarks: between +6.69 and +8.26 points on SPY (10 of 10 seeds positive in every budget) and between +3.89 and +6.44 points on QQQ (8–9 of 10 seeds positive). Budget medians differ by a few points with heavily overlapping interquartile ranges, so the evidence supports robustness of the candidate source to CEM random initialization rather than a preference for any optimizer budget. On the seed-42 flagship path, a 5-day moving-block bootstrap of daily excess return gives one-sided $p = 0.0317$ on SPY and $p = 0.3102$ on QQQ; given a single, short, and repeatedly inspected 111-day window, we treat this as descriptive rather than confirmatory.

7.4 Do the treatments beat the frozen baseline?

Table 6 answers the comparison the matrix cannot: paired on the same seed at the standard 6×20 budget, the all-treatment configuration *underperforms* the frozen batch-fit baseline. The median excess-return difference is −7.43 points on SPY (0 of 10 seeds favor the treated arm) and −3.44 points on QQQ (3 of 10). This difference bundles two changes — the switch from a frozen pre-2026 policy to an online walk-forward schedule, and the addition of T1, T3, and T4 — and the seed-42 decomposition in Table 4 suggests the protocol switch and the T3 sizing change account for much of it. The treated arm trades less and pays lower transaction costs, but its median Sharpe is also lower. The honest conclusion is that the studied treatments produce a profitable historical implementation without improving on the simplest frozen implementation; their value in this sample is architectural (label-complete chronology, bounded sizing, explainable allocation) rather than performance-based.

8 Discussion

The primary empirical result is a positive observed conditional mean in the cleaned opportunity set, economically large relative to modeled cost and stable under event-cluster resampling. It is

not established under every reasonable dependence model (symbol, week, and month clustering cross zero, as does the equal-event month block), it is upper-tail dependent, and no benchmark-relative interval excludes zero. The appropriate conclusion is a positive historical conditional expectation with limited effective sample size and substantial concentration, not a universal return law and not demonstrated market-relative excess.

The strongest additional result is heterogeneity, and it must be read jointly with concentration. Earnings questions contribute approximately zero mean return, while the geopolitical subset is positive in both mean and median on only 37 economic events — and removing one oil-linked symbol or one month removes most of the full-sample mean. This supports stricter candidate selection and motivated T4’s family-aware allocation, but it does not prove that geopolitical markets contain private information: a common public-news process, commodity exposure, recurring symbols, or favorable months can generate the same pattern, and the paired portfolio comparison shows the allocation rule did not convert the pattern into excess portfolio performance.

Polarity and cleaning serve different scientific roles: polarity makes the signal economically interpretable (its value is not judged by whether the historical mean rises), while cleaning defines the estimand. The matched CEM comparison is directionally favorable to cleaning but statistically inconclusive.

For live use, the defensible role of Polymarket is a continuous confirmation variable informing eligibility and exits; a prospective hourly implementation should preserve the same semantic rules and evaluate only data observable at each decision time.

The principal limitations are the two-year venue history, one prediction-market source, a targeted rather than universe-complete semantic audit, daily historical equity decisions, and eligibility parameters inherited from the portfolio optimizer rather than fixed a priori. The event endpoint is the scheduled value stored in the research dataset: endpoint revisions and actual resolution timestamps are not versioned, so it cannot be proven that every event was still unresolved at the measured exit close. Repeated research iterations mean that the 2026 period is not a pristine untouched holdout, and the walk-forward arms refit within it. The seed/budget protocol addresses optimizer-initialization luck, but only future data can provide a genuinely prospective test.

Table 5: Flagship T1+T2+T3+T4 distribution across ten seeds over the full January–June 2026 test period. Entries are median [25th, 75th percentile] excess return in percentage points, the count of seeds with positive excess, and the median maximum drawdown. No budget is selected or ranked by these results.

Budget	SPY			QQQ		
	Excess median [IQR]	> 0	MaxDD	Excess median [IQR]	> 0	MaxDD
6×20	+8.26 [+4.31, +9.63]	10/10	−8.74	+3.89 [+2.21, +7.25]	8/10	−9.26
6×30	+6.69 [+6.15, +8.37]	10/10	−8.73	+5.65 [+3.11, +7.69]	9/10	−9.48
10×20	+7.76 [+7.35, +10.52]	10/10	−8.62	+4.65 [+1.33, +5.96]	8/10	−9.39
10×30	+6.97 [+5.77, +9.06]	10/10	−8.47	+6.44 [+4.15, +7.23]	9/10	−9.54

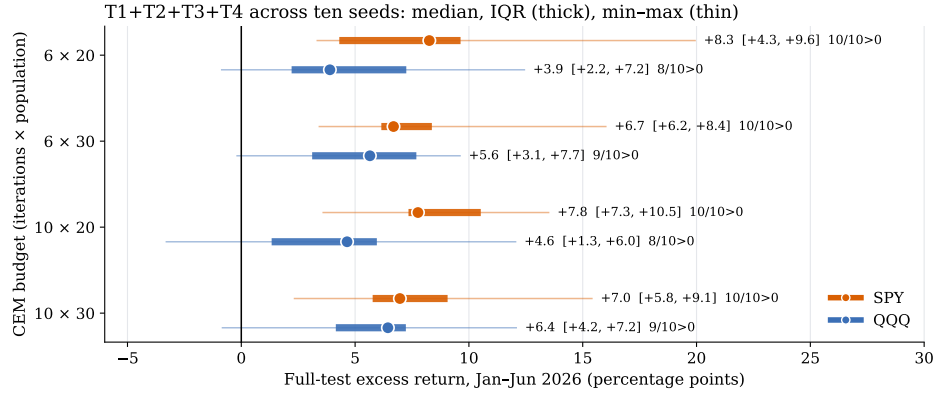


Figure 5: Flagship T1+T2+T3+T4 optimizer robustness across ten seeds and four CEM budgets over the full January–June 2026 test period. Medians are positive throughout; no budget is selected from these results.

Table 6: Paired comparison at the standard 6 × 20 budget: T1+T2+T3+T4 minus the frozen batch-fit Baseline on the same seed and benchmark (ten seeds each). Entries are median [IQR] differences over the full test period; Wins is the share of seeds where the treated configuration has higher excess return.

	Δ Excess [IQR]	Δ Sharpe	Δ MaxDD	Δ Cost (\$)	Wins
SPY	−7.43 [−12.39, −3.92]	−0.92	−0.16	−2,032	0%
QQQ	−3.44 [−7.87, +1.06]	−0.55	−1.00	−1,620	30%

9 Conclusion

Four findings stand. First, after exact duplicate collapse and targeted semantic cleaning, 887 valid Polymarket-linked candidate observations have mean net equity return +1.208% between policy-accepted entry and the final eligible pre-event close; the estimate weakens under symbol and calendar dependence and under upper-tail removal. Second, heterogeneity is real but concentrated: geopolitical observations dominate earnings observations, yet removing one oil-linked symbol (USO) or March 2026 alone removes most of the full-sample mean. Third, matched SPY and sector-ETF controls leave no 95% interval that excludes zero, so the benchmark-relative evidence does not establish statistically reliable excess return. Fourth, a fully specified CEM implementation converts the candidate source into positive historical portfolios across ten optimizer seeds and four

search budgets on one contiguous January–June 2026 test period — while the studied treatments do not outperform the frozen batch-fit baseline in seed-paired comparisons, and the short, repeatedly inspected test period does not establish prospective profitability.

The research contribution is therefore narrower and more defensible than information diffusion. Polymarket supplies continuous, timestamped crowd-belief variables that can identify event relevance, encode the favorable outcome through polarity, and confirm entry or exit conditions for selected equities. Establishing why the conditional return exists, whether it beats simple market exposure, and whether it persists prospectively requires new events rather than additional optimization of the same history.

References

- [1] F. A. Hayek. 1945. The Use of Knowledge in Society. *American Economic Review*.
- [2] E. F. Fama. 1970. Efficient Capital Markets. *Journal of Finance*.
- [3] S. J. Grossman and J. E. Stiglitz. 1980. On the Impossibility of Informationally Efficient Markets. *American Economic Review*.
- [4] J. Wolfers and E. Zitzewitz. 2004. Prediction Markets. *Journal of Economic Perspectives*.
- [5] J. E. Berg, F. D. Nelson, and T. A. Rietz. 2008. Prediction Market Accuracy in the Long Run. *International Journal of Forecasting*.
- [6] D. Duffie. 2010. Asset Price Dynamics with Slow-Moving Capital. *Journal of Finance*.
- [7] B. Cowgill and E. Zitzewitz. 2015. Corporate Prediction Markets. *Review of Economic Studies*.
- [8] V. L. Bernard and J. K. Thomas. 1989. Post-Earnings-Announcement Drift. *Journal of Accounting Research*.